

School of Engineering

Diploma in AI & Data Engineering

EG2A17 — Data Engineering Project WIU

Final Report · AY2026 Semester 1

NESSO SAFETY MONITOR

A Real-Time Data Engineering Pipeline for Fall and Near-Miss Detection on Construction Sites

EG2A17 – A4|Group 5

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1. Project Team Members

S/N	LU Group	Name	Admin No.	Role and Responsibilities
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2. Executive Summary

Falls remain a serious construction-site hazard, and delayed awareness can slow intervention. This project delivers the Nesso Safety Monitor: a local-first data engineering pipeline that acquires six-axis inertial data from Nesso N1 wearables over Bluetooth Low Energy (BLE), processes the stream in near real time, stores structured safety events, and presents worker status, live signals and incident history through a Streamlit dashboard. The implemented scope covers acquisition, cleaning, feature engineering, rule-based detection, storage, retrieval, visualisation and multi-worker live operation.

Four Nesso N1 devices were configured at a nominal 25 Hz. Twelve labelled recordings from four wearers across normal-work, near-miss and controlled fall-like activities produced 35,221 validated samples (≈ 23.5 minutes). Six raw axes were reduced to four scalar features—acceleration magnitude, gyroscope magnitude, jerk and rotation activity. The validated Stage-1 rule treats any 2 of 4 threshold exceedances as a SAFETY

EVENT; the prototype UI then displays 0–1 votes as SAFE, 2–3 as NEAR-MISS and 4 as FALL. The severity display is indicative rather than a separately validated fall-versus-near-miss classifier.

Leave-one-worker-out cross-validation detected 7 of 7 reliable, labelled incidents with zero false negatives. The separately flagged Ziqian fall-like recording remained an uncertain stress test and was not detected, while 1 of 4 normal-work recordings produced an alert. These results show that the end-to-end pipeline is functional, but the operating point is not production ready. Stored live-event data also showed that acceleration and jerk alone do not cleanly separate the prototype severity labels; rotation activity provided the strongest separation in this small sample.

3. Introduction

3.1 Background and Problem Statement

Singapore's workplace safety performance is strong internationally, but construction remains the sector contributing the largest share of serious harm. The Ministry of Manpower reported 43 workplace fatalities in 2024, of which construction accounted for 20, and the sector remained the top contributor to combined fatal and major injuries at 166 cases (MOM, 2025). Within construction in 2024 the team identified 44 fall-from-height fatal and major injuries and 29 slips, trips and falls. Two characteristics make these incidents hard to respond to: sites are large, noisy and visually obstructed, and work is often performed alone or in small, dispersed teams. A fallen worker may therefore lie undiscovered — the incident is instantaneous, but awareness of it is not.

A fall produces a distinctive inertial signature — an impact transient, rapid angular reorientation, and post-impact inactivity — that a wearable inertial measurement unit (IMU) can capture within a fraction of a second. The engineering problem is therefore not whether the signature exists, but how to build a pipeline that captures it, distinguishes it from vigorous ordinary work, stores it durably and puts it in front of a person fast enough to matter:

How can we automatically detect fall-from-height, slip-trip-fall and near-miss events, and alert authorised personnel for prompt intervention, to improve construction-site safety?

3.2 Objectives and Scope

Objective	Acceptance criterion	Status
O1 — MONITOR worker movement in real time using the Nesso N1	Six-axis IMU data is acquired over BLE, timestamped by the local python gateway and logged at a nominal sampling rate of 25 hz.	Fulfilled at prototype scale
O2 — DETECT abnormal movement and classify SAFE / NEAR-MISS / FALL by rule-based analysis	A frozen four-signal threshold configuration identifies safely events using a 2-of-4 voting rule, with the feature peaks and vote count retained for review.	Fulfilled at prototype scale
O3 — ALERT supervisors with live status, sensor graphs and event alerts via a web dashboard	A local Streamlit dashboard displays worker/device status, live sensor movement and safely-event history retrieved from the local data stores.	Fulfilled locally; external SMS/ push alerting not implemented

The prototype is a local-first worker-safety data pipeline using NESSO N1 IMU acquisition, BLE, a Python gateway, data cleaning/feature engineering, an explainable four-signal voting detector, CSV/SQLite storage, and a local Stream lit dashboard. It can collect, detect, and store data without cloud or Internet access. The detector uses acceleration magnitude, gyroscope magnitude, jerk, and rotation activity; ≥ 2 signals = SAFETY EVENT, while the prototype UI maps 0–1 = SAFE, 2–3 = NEAR-MISS, 4 = FALL. This mapping is not a separately validated fall-vs-near-miss classifier; only the 2-of-4 safety-event trigger is validated.

Outside the validated scope are real fall-from-height testing, public/cloud dashboard deployment, SMS/push alerts, worker localisation, and a production ML classifier; all fall-like tests were controlled floor-level movements for safety. The prototype assumes reasonably consistent NESSO placement, recognises that four team members do not represent the full construction workforce, and uses 25 Hz because testing showed a practical balance between motion capture, BLE stability, processing load, and storage volume—not because 25 Hz is claimed to be universally optimal for fall classification.

4. Methodology

4.1 System Architecture and Pipeline

The NESSO Safety Monitor is a local-first, multi-worker data pipeline covering data collection, processing, detection, storage, retrieval and visualisation. Each NESSO N1 samples three accelerometer and three gyroscope axes at a nominal 25 Hz and transmits the six-axis IMU data over BLE. The system does not require cloud or Internet connectivity for live operation, making it suitable for construction environments with unreliable network coverage.

The Python gateway validates BLE packets and worker/device identity, computes acceleration magnitude, gyroscope magnitude, jerk and rotation activity, and applies the frozen 2-of-4 safety-event rule. Each worker has an independent BLE session and detector state, so one worker's event or disconnection does not affect another. Recent samples are temporarily stored in SQLite for the live dashboard, while worker status, detector configuration and permanent events are stored for retrieval and analysis. CSV recordings are

retained separately for offline cleaning, calibration and validation. The Streamlit dashboard displays live worker status, sensor signals, thresholds and incident history.

4.2 Data Collection and Cleaning

Four team members acted as workers, giving four independent gait patterns rather than a single-subject dataset. Each performed three separately recorded classes: normal work (walking, carrying, bending, stairs), near-miss (a slip or trip recovered without reaching the ground), and fall-like (a controlled descent at floor level). Twelve CSV recordings were captured automatically, each carrying a timestamp, elapsed seconds, measured inter-sample interval and the six axes, with the time of the deliberate event noted so detections could be verified against a known ground-truth location rather than merely counted.

Cleaning enforced a single schema contract in five idempotent steps: standardise variant column names onto the canonical six-axis schema; validate that all six required columns exist, rejecting a non-conforming file outright rather than processing it partially; coerce sensor values to numeric and drop rows with missing required values; remove duplicate and corrupted records; and sort by time, label by worker and activity, and concatenate into one combined dataset. The pipeline emits a per-file audit — rows before, rows after, and rows removed by cause — which converts cleaning from an unverifiable step into an evidenced one.

4.3 Feature Engineering

Raw per-axis values depend strongly on how the device happens to sit on the body, so the six axes are reduced to four orientation-tolerant scalar features.

Feature	Definition	Unit	Physical meaning
Acceleration magnitude, A	$A = \sqrt{Xg^2 + Yg^2 + Zg^2}$	g	Overall movement and impact force, independent of orientation
Gyroscope magnitude, G	$G = \sqrt{Xdeg^2 + Ydeg^2 + Zdeg^2}$	°/s	Overall angular rate; a fall reorients the body rapidly, walking does not (degree per second)
Jerk, J	$J = \Delta A / \Delta t$	g/s	How abruptly motion changes; separates a sudden impact from a firm gradual movement of similar peak
Rotation activity, R	$R = \int G \, dt$ over the window	°	Accumulated angular travel across the event window

For offline recordings, jerk uses the measured inter-sample Δt where available, and rotation activity integrates gyroscope magnitude over the event window. The lesson-ready BLE payload does not include a device timestamp, so the live gateway reconstructs elapsed time from the fixed 25 Hz source cadence ($\Delta t = 0.04$ s) rather than bursty BLE arrival timing. Rotation activity therefore represents accumulated angular travel, not net body angle.

4.4 Threshold Calibration and Detection Rule

The four features were compared across normal and abnormal recordings to determine suitable thresholds. The final thresholds were fixed at 3.281536 g for acceleration, 641.286497 dps for gyroscope, 67.407807 g/s for jerk, and 524.512582° for rotation activity. Each feature that exceeds its threshold contributes one vote, and a minimum of 2 out of 4 votes triggers a SAFETY EVENT. The dashboard displays 0–1 votes as SAFE, 2–3 as NEAR-MISS and 4 as FALL. However, only the 2-of-4 safety-event detection rule was validated; the NEAR-MISS and FALL labels are prototype classifications.

Votes (of 4 features)	Classification	Interpretation
0 – 1	SAFE	Ordinary work activity, including vigorous but unremarkable movement
2-3	NEAR-MISS	Disturbance with rotation — consistent with a slip or trip that was recovered
4	FALL	Impact with rotation and abrupt acceleration change — consistent with a fall

The detection rule is designed to be auditable and configurable. Each stored safety event records the total vote count together with the peak acceleration, gyroscope, jerk and rotation values. These values can be compared with the active threshold configuration to reconstruct why an event was triggered. The `detector_config` table also stores three operational parameters: a 5 s startup grace period to prevent false detections while the device stabilises, an 8 s cooldown to prevent the same physical event from being logged repeatedly, and the required-votes setting used to control detector sensitivity. Keeping these parameters in the configuration rather than embedding them only in the detection logic also makes future retuning easier and keeps historical events interpretable. The current event schema does not store the four individual feature-vote bits, so persisting this vote vector would further improve traceability in future versions.

4.5 Data Management System

Storage is separated according to its purpose. CSV files are retained for offline analysis and validation, while SQLite is the system of record during live operation. Recent sensor readings are stored temporarily in the `live_samples` table to support the live dashboard and are automatically removed after the retention period so that high-rate sensor data does not grow indefinitely. Permanent safety events and system state remain stored in structured SQLite tables.

The database contains six application tables: `workers` for worker/device identity, `device_status` for current connection and safety state, `events` for permanent safety-event history, `detector_config` for versioned thresholds and detector parameters, `offline_queue` to support future retry/buffering workflows, and `live_samples` as a rolling buffer for dashboard visualisation. Live samples are written in batches to reduce unnecessary database operations. Each event is validated against the registered worker and device before storage, and every event has a unique event_uuid to prevent duplicate ingestion. Database constraints also restrict invalid event values, while the versioned `detector_config` keeps historical events interpretable if thresholds are changed in future. This design provides structured storage, efficient live retrieval and traceability while remaining appropriate for the prototype's local-first architecture.

4.6 Validation Methodology

Validation used leave-one-worker-out cross-validation on the fixed rule design: each worker was held out in turn while thresholds were derived from the other workers' normal data, then the held-out incidents and normal activity were tested. The uncertain Ziqian fall-like recording was excluded from the reliable incident score and retained as a stress test. After selecting the rule, the final full-data thresholds were frozen and replayed end to end; startup grace and cooldown were then tuned with zero false negatives on the reliable incidents as the first priority.

The resulting figures are prototype evidence, not a claim of production accuracy. The reliable set contains only seven scored incidents from four similar participants, one uncertain fall-like stress test was missed, and the final operating point still produced a normal-work false alarm. Independent site data and substantially greater participant diversity are required before deployment claims are justified.

5. Results and Findings

5.1 Dataset and Data Quality

The twelve recordings yielded 35,221 validated samples after cleaning, equivalent to approximately 1,409 seconds of continuous six-axis motion data. The class distribution is close to balanced, which matters because a heavily skewed dataset would make the detector appear more accurate simply by favouring the majority class.

Movement class	Recordings	Samples	Share	≈ Duration
Normal work	4	14,157	40.2%	566 s
Near-miss	4	10,463	29.7%	419 s
Fall-like	4	10,601	30.1%	424 s
Total	12	35,221	100%	≈1,409 s

Finding 1 — the acquisition path produced structurally complete data. Across all twelve recordings the cleaning pipeline removed zero rows: no missing required values, no duplicates, no corrupted readings. This is a positive result about the capture chain, not evidence that cleaning was unnecessary. Its value here was standardisation and verification — reconciling inconsistent column naming across four capture sessions, enforcing the schema contract, and producing auditable proof that the data entering analysis was complete. A pipeline that cannot demonstrate this is one whose downstream results cannot be trusted.

Finding 2 — the measured sampling interval varied around the nominal 40 ms period. Recorded inter-sample intervals ranged from approximately 29 ms to 48 ms, showing that BLE arrival timing is not perfectly uniform even though the average throughput remains close to the intended 25 Hz. This matters particularly for jerk, because using an incorrect time interval would directly affect the calculated value. For offline recordings, the measured inter-sample interval is therefore used where available. In the live gateway, the

BLE payload does not contain a device timestamp and packet arrival can be bursty, so feature computation uses the fixed 25 Hz source cadence ($\Delta t = 0.04$ s). This keeps the live detector consistent with the configured sensor sampling rate rather than treating BLE arrival jitter as worker movement.

5.2 Detection Performance

The frozen final configuration detected all 7 reliable, labelled incidents and produced 0 false negatives on that scored set. The uncertain Ziqian fall-like stress test was not detected and is reported separately rather than hidden inside the recall figure. Among the four normal-work recordings, one produced at least one alert. At the scanned-position level, the final thresholds produced a 0.46% normal-position false-positive rate; at recording level, the normal false-alarm rate was 1 of 4 (25%).

Ground truth	Detector alerted	Detector silent	Metric	Value
Reliable incident (n = 7)	7	0	Recall	7/7 = 100%
Normal recording (n = 4)	1	3	Recording false-alarm rate	1/4 = 25%
Normal scan positions	-	-	Position false-positive rate	0.46%
Uncertain Ziqian fall-like stress test (n = 1)	0	1	Scored separately	Not in recall

Interpretation of the false positive. A safety system must avoid both missed incidents and nuisance alarms. One of four normal recordings triggered, so the current operating point is not ready for production, even though the per-position false-positive rate is low. The dataset is too small to extrapolate a credible hourly nuisance-alarm rate; the correct conclusion is that more normal-work data and harder negative examples are needed before sensitivity is increased or deployment is considered.

5.3 Anomaly and Trend Analysis

Because each stored event contains the vote count and peak values of the four detector features, the events table can support analysis in addition to real-time monitoring. Eleven live-operation events recorded on 14 August—ten displayed as NEAR-MISS and one as FALL by the prototype UI—

were compared to investigate how the four features differed between the displayed severity states. These labels are outputs of the prototype vote mapping and are not independently verified fall-versus-near-miss ground truth.

Feature	UI NEAR-MISS events (n = 10)	UI FALL event	Comparison
Peak acceleration	Mean 4.53g, range 2.92 - 5.71g	5.95g	Higher, but only 4.2% above the highest near-miss
Peak gyroscope	Mean 817dps, range 542-1123 dps	1131dps	Higher, but only 0.7% above the highest near-miss
Peak jerk	Mean 55.4 g/s, range 34.3-76.8g/s	68.1g/s	Overlaps; a near-miss recorded higher jerk
Rotation activity	Mean 317°, range 205–563°	658°	Strongest separation: 16.9% above the highest and about 2× the mean

Principal finding. The four features did not contribute equally to the prototype severity display. Peak acceleration and gyroscope were slightly higher for the UI FALL event than for all ten UI NEAR-MISS events, but the margins were small. Jerk showed clear overlap, as one near-miss event recorded a higher value than the fall event. Rotation activity showed the strongest separation, with the UI FALL value approximately 16.9% above the highest near-miss value and more than twice the near-miss mean. This supports the use of multiple movement features rather than relying on a single sensor measurement.

However, with only one UI FALL event and prototype-generated labels, these results are exploratory and do not establish a validated fall-versus-near-miss classifier.

Trend and audit finding. Nine of the eleven events occurred within approximately two minutes. Because these events came from concentrated system testing, this clustering should not be interpreted as a real construction-site safety trend. However, it demonstrates that the database structure can support temporal and worker-level analysis. Queries using `event_time`, `worker_id`, `event_votes` and the stored feature peaks could identify repeated alerts for a worker, periods with unusually high event frequency, or changes in event severity over a longer deployment. Location-based hotspot analysis is not currently possible because worker localisation has not been implemented.

The stored feature peaks and aggregate vote count also provide useful audit information. Under the current prototype mapping, 0–1 votes = SAFE, 2–3 votes = NEAR-MISS and 4 votes = FALL, so the previously observed three-vote near-miss is consistent with the implemented logic rather than evidence of a classification error. A future improvement would be to store the four individual feature-vote bits together with the aggregate vote count, making it immediately visible which thresholds contributed to each event without requiring the decision to be reconstructed from the feature values.

5.4 Delivered System and Limitations

The dashboard provides an Overview with headline counts and worker cards, a live signal view, an Incident Log, and a Detector view exposing the active thresholds. Connection freshness is checked so a silent device is shown as disconnected rather than safe. Live samples are queried from SQLite, while permanent events retain `event_time`, `received_at`, `worker/device` identity, vote count and feature peaks for later review.

Four limitations bound the results. (i) Only four similar participants and seven reliable, scored incidents were available, so generalisation is uncertain despite leave-one-worker-out cross-validation. (ii) No real fall from height was tested; all fall-like movements were controlled at floor level. (iii) The uncertain Ziqian fall-like stress test was missed, proving the current rule does not cover every abnormal pattern. (iv) One local gateway and SQLite database remain a single point of failure and are appropriate only at prototype scale.

6. Alternative Solutions and Future Enhancements

6.1 Alternative 1 — Edge-Side Detection on the Wearable

Moving feature computation and thresholding onto the Nesso N1 would let each device transmit only classified events and a heartbeat. It would decide within one sample period rather than after a BLE hop, cut airtime and battery drain by orders of magnitude, and, decisively, keep detecting outside BLE range by buffering on-device until reconnection. Against it: threshold changes would need a firmware or configuration-push channel rather than one database row, and the raw trace would be lost unless a pre-event buffer is uploaded. Gateway-side detection was correct here because retaining the raw signal was essential to calibration and to the analysis in Section 5.3; for production the coverage argument wins, since a safety device that stops detecting when it loses radio contact fails precisely where a fall is most dangerous. The recommended target is a hybrid — edge detection with a short raw buffer uploaded alongside each event.

6.2 Alternative 2 — Supervised Machine Learning in Place of Fixed Thresholds

A classifier trained on windowed features would replace the fixed-threshold vote. Its decisive advantage is that it learns feature interactions, and Section 5.3 proves such interactions exist — the four features are neither independent nor equally informative, which is exactly what an equal-weight vote cannot express. A learned boundary could also suppress the specific normal-work pattern causing the false positive without uniformly desensitising the detector, whereas raising a threshold risks suppressing genuine low-energy events. Against it: twelve recordings from four similar wearers would over-fit badly; explainability in an incident investigation would need extra tooling, where the present rule reduces to which features fired and by how much; and model versioning and drift monitoring are a far larger operational commitment. Machine learning is the right direction but the wrong immediate step, because the binding constraint is data diversity, not model capacity.

Recommended sequence: (1) re-validate the current rule under leave-one-worker-out on existing data; (2) add a post-event quiescence feature — sustained low acceleration variance after impact — which is the clearest visual discriminator between a recovered near-miss and a completed fall and targets the false-positive mode directly; (3) weight the votes rather than treating them equally, giving rotation activity greater influence; (4) collect a substantially larger and more varied labelled dataset; (5) only then train a classifier, retaining the rule-based detector as an always-on fallback so a model failure cannot leave a worker unmonitored.

6.3 Alternative 3 — Server-Based Streaming Store in Place of Local SQLite

MQTT feeding a server-side relational or time-series database would replace local SQLite as the system of record. It supports many concurrent producers, native multi-site aggregation, and first-class retention, compression and downsampling for the high-rate log that currently grows without bound. Against it: it requires broker, server, backup, monitoring and access control, and without connectivity it degrades to local buffering — the offline queue reimplemented at greater cost. SQLite was correct at prototype scale and should be retained as the on-gateway buffer regardless, because local-first durability keeps the system honest when the network is unavailable. The migration trigger is concrete: move when a second gateway is introduced, because that is where the single-writer model stops being a simplification and becomes a defect. A staged path — SQLite as

local buffer with replication forwarding events upstream — preserves offline operation without rewriting the ingestion path.

6.4 Future Enhancement 1 — Indoor Location Tracking

The dashboard answers who and what, but not where; on a multi-storey site that is the difference between knowing a worker has fallen and being able to reach him. Response time is the outcome the system exists to improve, and detection without location addresses only its first half. The proposal trilaterates BLE received signal strength against three or more fixed anchors at surveyed points, overlaid on a site plan, with expected accuracy of $\pm 2\text{--}5$ m indoors — enough to resolve a zone or floor, the operationally useful granularity. The pipeline impact is additive: an anchor-readings table and a position estimate joined to the existing event record, detection path unchanged. The principal risk is that RSSI is highly sensitive to multipath, metal structure and human bodies, all abundant on a site, so accuracy must be validated per site, and the dashboard must show an uncertainty radius rather than a false-precision point.

6.5 Future Enhancement 2 — Escalating Out-of-Band Alerting

Alerting is confined to the dashboard, which assumes someone is watching a screen — an assumption that fails at the moment it matters, and the design cannot detect that an alert has gone unseen. The proposal adds an escalation ladder with mandatory acknowledgement: dashboard banner and audible alert at T+0; push or SMS to the on-duty supervisor at T+30 s unacknowledged; escalation to the site safety officer and optional siren at T+2 min. The events table already carries acknowledged and acknowledged_at, so the audit trail exists and only the workflow and timer are missing. Time-to-acknowledge then becomes a measurable safety KPI which, with the near-miss trend analysis of Section 5.3, gives a site a leading rather than lagging metric. This enhancement should only be introduced after the false-positive behaviour identified in Section 5.2 is reduced. Repeated nuisance alerts could create alarm fatigue, reduce supervisor trust and eventually cause genuine alerts to be ignored. Therefore, improving detector reliability should precede automatic escalation.

7. Conclusion

The NESSO Safety Monitor delivers a complete local-first data engineering prototype for construction-site safety monitoring. Four NESSO N1 wearables collect six-axis IMU data at 25 Hz and send it through BLE to a Python gateway, where the data is validated, transformed into four motion features, analysed using a frozen 2-of-4 voting rule, stored in SQLite and displayed through a Streamlit dashboard.

The final detector identified all 7 reliable, labelled safety events with zero false negatives. However, 1 of 4 normal-work recordings produced a false alert, while an additional uncertain fall-like stress test was not detected and was reported separately. Stored event analysis also showed that rotation activity provided the strongest separation between the prototype UI states, although the severity labels were not independently validated.

Overall, the project demonstrates an integrated pipeline covering acquisition, processing, detection, storage, retrieval, visualisation and analysis. Future work should focus on collecting more diverse data, reducing false

alarms, improving traceability and extending the system towards more scalable multi-gateway or edge-based deployment.

8. References

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9. Appendices

Supporting evidence — validation CSV, per-file cleaning audit, SQLite DDL, gateway and dashboard source, and the presentation deck — is submitted alongside this report, which ran to the Week 11–18 schedule. Appendix A records the frozen configuration needed to reproduce every result above.

Appendix A — Frozen detector configuration. Thresholds: acceleration ≥ 3.281536 g; gyroscope ≥ 641.286497 dps; jerk ≥ 67.407807 g/s; rotation activity $\geq 524.512582^\circ$. Required votes: 0–1 SAFE, 2–3 NEAR-MISS, 4 FALL; ≥ 2 = SAFETY EVENT. Startup grace and cooldown are configured per deployment. Calibration and validation dataset: 12 recordings, 4 wearers, 3 movement classes, 35,221 rows.

10. Acknowledgement

AI tools, including